On a re-scaled rough Heston model with jumps - weak convergence of the log stock price to a Lévy marginal

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Abstract

We show weak convergence of the marginals for a re-scaled rough Heston model to the marginal of a Normal Inverse Gaussian (NIG) Lévy process. This shows we can obtain such a limit without having to impose that the true Hurst exponent H for the model is $\frac{1}{2}$ as in [AC24],[1] or that $H \searrow -\frac{1}{2}$ as in [AAR25], so the result potentially has increased financial relevance. We also extend to the case when V has jumps, where we show weak convergence of the log stock price marginals to the marginal of an arithmetic Brownian motion evaluated at the inverse hitting time process of a linear combination of the same Brownian plus another independent Brownian motion with drift plus an independent Lévy process.

1 Introduction

The Rough Heston stochastic volatility model was introduced in [JR16], and (using $C-$tightness arguments) they show that the model arises naturally as a weak large-time limit of a high-frequency market microstructure model driven by two nearly unstable Hawkes processes. [ALP19],[ER19] and [GK19] show that the characteristic function of the log stock price for Rough Heston-type models can be expressed in terms of the solution to a non-linear Volterra integral equation (VIE) (see also [ER18] and [BLP24],[BPS24], for extensions to jumps in V), which allows for accurate option pricing for $H \ll 1$ (and even $H = 0$) using an Adams scheme to solve the VIE numerically. This avoids Monte Carlo (MC) techniques for which traditional MC schemes are notoriously inaccurate for $H \ll 1$ (both in terms of bias and sample variance). From the stochastic Fubini theorem, it is well known that $A_t = \int_0^t V_s ds$ satisfies an equation of the form: $A_t = G_0(t) + \int_0^t \kappa(t-s) W_{A_s} ds$ for some Brownian motion W (see e.g. Theorem 3.1 in [JR20]), which is a non-linear VIE for A in terms of the a.s. $(\frac{1}{2} - \varepsilon)-$Hölder continuous function $W_{(\cdot)}$, and is still well defined even if κ is only in $L^1(0,T)$; this also allows us to consider the hyper-rough regime $H \in (-\frac{1}{2}, 0]$. If we discretize this SVE and re-write in terms of the final (discrete-time) increments of A and W, then we can use an independent sequence of Inverse Gaussian random variables to perform an approximate Monte Carlo simulation of A (see Algorithm 1 in [AA25] for details) which is naturally suited to the regimes $H \ll 1$ and $H \in (-\frac{1}{2}, 0]$. The variance process V is $(H - \varepsilon)-$Hölder continuous like fBM (see e.g. Theorem 3.2 in [JR16]) and the spot and VIX smiles exhibit power-law skew for implied volatility in the small-time limit (see [FGS21], [FGS22] for details). See also [ALP19], [Cuch22], [FGS21], [FS21] for further related results on rough Heston-type models.

[BL24] improve accuracy in estimating the tail part of the integral for Fourier inversion (for e.g. call options under the rough Heston model) using a sinh contour which goes outside the usual strip of analyticity for the mgf (but is admissible as long as it avoid poles of the characteristic function), and also discuss refinements to the usual Adams schemes for solving the rough Heston VIE.

[AC24] show that a re-scaled standard Markov Heston model with fast mean-reversion and large vol-of-vol (via an H parameter which is not the Hurst exponent) tends weakly on path space to one of three different models (either Black-Scholes, a Normal Inverse Gaussian or a Normal Lévy model), depending on whether their H parameter is $>$, $=$, or $< -\frac{1}{2}$. [AAR25] obtain a similar result without any ε parameter but instead letting $H \searrow -\frac{1}{2}$ for the so-called *hyper-rough Heston model* (see also Section 5 in [FGS21] and Section 7 in [AJ21] for more on this model), and exploiting Dirac-type behaviour in their Lemma 2.4.

In this note, we fill in the gap between [AC24] and [AAR25], by showing that a conceptually similar result is obtained for any $H \in (0, \frac{1}{2})$ (in particular our regime for $H = \frac{1}{2}$ corresponds to the regime in Eq. (0.3) in [AC24] with their $H = -\frac{1}{2}$). We later extend the model to allow positive jumps in V^ε, and in this case (using the Laplace transform of the hitting time to a lower barrier for a spectrally positive Lévy process), we find that the limiting process for the integrated variance is a deterministic time-change of the first passage time process to lower barriers for a more general class of Lévy processes.

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[1] The H parameter in [AC24] is not fixed to be $\frac{1}{2}$, but the Hurst exponent for their family of V^ε processes is of course $\frac{1}{2}$ since the models are all standard Heston for $\varepsilon > 0$.

2 Asymptotics for the terminal log stock price for a re-scaled rough Heston model

We work on a probability space $(\Omega, \mathcal{F}, \mathbb{Q})$ throughout with filtration $(\mathcal{F}_t)_{t\geq 0}$ satisfying the usual conditions.

Proposition 2.1 *Consider a re-scaled rough Heston model for a log stock price process X_t^ε:*

$$\begin{aligned} dX_t^\varepsilon &= -\frac{1}{2}V_t^\varepsilon dt + \sqrt{V_t^\varepsilon}dB_t, \\ V_t^\varepsilon &= V_0 + \frac{1}{\Gamma(\alpha)}\int_0^t (t-s)^{\alpha-1}(\frac{1}{\varepsilon}\lambda(\theta - V_s^\varepsilon)ds + \frac{1}{\varepsilon}\sigma\sqrt{V_s^\varepsilon}dW_s), \end{aligned} \tag{1}$$

where B and W are two Brownian motions with $dB_t dW_t = \rho dt$ with $\rho \in [-1,0]$, $\alpha \in (\frac{1}{2}, 1)$ and $\lambda, \sigma > 0$. Then (for $t > 0$ fixed) X_t^ε tends weakly to X_t as $\varepsilon \to 0$, where X_t has the marginal law of a Normal Inverse Gaussian Lévy process which does not depend on $H = \alpha - \frac{1}{2}$ or V_0.

Proof. Let $I^\alpha(f)(t) = \frac{1}{\Gamma(\alpha)}\int_0^t (t-s)^{\alpha-1}f(s)ds$ denote the αth-order fractional integral of a function f for $\alpha \in (0,1]$, and (without loss of generality), we assume $X_0^\varepsilon = 0$. Then for $p \in (0,1)$ (which will be sufficient for our purposes when we invoke the [Bill86] weak convergence result below)

$$\mathbb{E}[e^{pX_t^\varepsilon}] = e^{V_0 I^{1-\alpha}\phi_\varepsilon(t) + \frac{1}{\varepsilon}\lambda\theta I^1\phi_\varepsilon(t)} \tag{2}$$

where ϕ_ε is the unique solution of the non-linear Volterra integral equation (VIE):

$$\phi_\varepsilon(t) = \frac{1}{\Gamma(\alpha)}\int_0^t (t-s)^{\alpha-1}(\frac{1}{2}(p^2-p) + \frac{1}{\varepsilon}(\rho p\sigma - \lambda)\phi_\varepsilon(s) + \frac{1}{\varepsilon^2}\frac{1}{2}\sigma^2\phi_\varepsilon(s)^2)ds$$

(see e.g. Section 4 in [ER19] or Section 7 in [ALP19]). Now let $\phi_\varepsilon(t) = \varepsilon\psi(\varepsilon^q t)$. Then

$$\begin{aligned} \varepsilon\psi(\varepsilon^q t) &= \frac{1}{\Gamma(\alpha)}\int_0^t (t-s)^{\alpha-1}(\frac{1}{2}(p^2-p) + (\rho p\sigma - \lambda)\psi(\varepsilon^q s) + \frac{1}{2}\sigma^2\psi(\varepsilon^q s)^2)ds \\ &= \frac{1}{\Gamma(\alpha)}\int_0^{\varepsilon^q t} (t - u\varepsilon^{-q})^{\alpha-1}(\frac{1}{2}(p^2-p) + (\rho p\sigma - \lambda)\psi(u) + \frac{1}{2}\sigma^2\psi(u)^2)du\,\varepsilon^{-q} \\ &= \frac{\varepsilon^{-q(\alpha-1)}}{\Gamma(\alpha)}\int_0^{\varepsilon^q t} (\varepsilon^q t - u)^{\alpha-1}(\frac{1}{2}(p^2-p) + (\rho p\sigma - \lambda)\psi(u) + \frac{1}{2}\sigma^2\psi(u)^2)du\,\varepsilon^{-q} \end{aligned}$$

where we set $\varepsilon^q s = u$ in the second line, so $\varepsilon^q ds = du$. Then setting $\varepsilon^q t \mapsto t$, we see that

$$\varepsilon\psi(t) = \frac{\varepsilon^{-q\alpha}}{\Gamma(\alpha)}\int_0^t (t-u)^{\alpha-1}F(\psi(u))du \tag{3}$$

where $F(w) = \frac{1}{2}(p^2-p) + (\rho p\sigma - \lambda)w + \frac{1}{2}\sigma^2 w^2$. If now we let $q = -\frac{1}{\alpha}$, the VIE (3) is independent of ε, so

$$\phi_\varepsilon(t) = \varepsilon\psi(\frac{t}{\varepsilon^{\frac{1}{\alpha}}}).$$

Hence for every $t > 0$ (from Lemma 4.4 in [FGS21])

$$\lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\phi_\varepsilon(t) = \lim_{t\to\infty}\psi(t) = \psi(\infty) = U_1(p) := \frac{1}{\sigma^2}[\lambda - p\sigma\rho - \sqrt{\lambda^2 - 2\lambda\rho\sigma p + \sigma^2 p(1-p\bar{\rho}^2)}]$$

for $p \in (0,1)$, where $\psi(\infty)$ is the smallest root of F and $\bar{\rho}^2 = 1-\rho^2$, and the convergence to $\psi(\infty)$ is rapid for $\varepsilon \ll 1$ since $\frac{1}{\varepsilon}\phi_\varepsilon(t) = \psi(\frac{t}{\varepsilon^{\frac{1}{\alpha}}})$. More precisely, Lemma 4.5 in [FGS21] can be applied to $-\psi$, which solves the following VIE:

$$-\psi(t) = \frac{1}{\Gamma(\alpha)}\int_0^t (t-u)^{\alpha-1}F_1(-\psi(u))du, \quad \text{with } F_1(w) = \frac{1}{2}(p-p^2) + (\rho p\sigma - \lambda)w - \frac{1}{2}\sigma^2 w^2$$

on $\mathbb{R}_+$, since $F_1(0) > 0$ and F_1 is analytic and decreasing on $\mathbb{R}_+$, the aforementioned result in [FGS21] implies that $-\psi(t)$ converges as $t \to \infty$ to the positive root of F_1, which coincides with $-U_1(p)$. Then for the exponent in (2), we know that $\frac{1}{\varepsilon}\phi_\varepsilon(.) = \psi(\frac{.}{\varepsilon^{\frac{1}{\alpha}}})$ is continuous on $\mathbb{R}_+$ and admits finite limit at ∞, and thus is bounded, so (by the dominated convergence theorem) we see that

$$V_0 I^{1-\alpha}\phi_\varepsilon(t) + \frac{\lambda\theta}{\varepsilon}I^1\phi_\varepsilon(t) = \frac{V_0}{\Gamma(1-\alpha)}\int_0^t (t-s)^{-\alpha}\phi_\varepsilon(s)ds + \lambda\theta\int_0^t \frac{1}{\varepsilon}\phi_\varepsilon(s)ds \to 0 + \lambda\theta U_1(p)t$$

as $\varepsilon \to 0$, and $\lambda\theta U_1(p)t$ is the log mgf of a Normal Inverse Gaussian Lévy process (see e.g. Remark 2.3 in [FJ11]). Then from e.g. the solution to Problem 30.4 on Page 573 in [Bill86] (which is also used in [GK19]), X_t^ε tends weakly to the marginal law of an NIG process. ■

2.1 Adding jumps into the V^ε process

We now extend the model to the following:

$$V_t^\varepsilon \;=\; V_0 + \frac{1}{\Gamma(\alpha)}\int_0^t (t-s)^{\alpha-1}(\frac{1}{\varepsilon}\lambda(\theta - V_s^\varepsilon)ds + \frac{1}{\varepsilon}\sigma\sqrt{V_s^\varepsilon}dW_s + \frac{1}{\varepsilon}d\tilde{J}_s^\varepsilon) \tag{4}$$

where $\tilde{J}^\varepsilon = (\tilde{J}_t^\varepsilon)_{t\geq 0}$ is given by

$$\tilde{J}_t^\varepsilon \;=\; \int_0^t\int_{\mathbb{R}_+} x(N^\varepsilon(dx,ds) - V_s^\varepsilon\nu(x)dxds)$$

and $N^\varepsilon(dx,dt)$ is a (time-inhomogeneous) Poisson random measure with (random) intensity $V_t^\varepsilon\nu(x)dxdt$; ν only has positive support with $\nu(\{0\}) = 0$ and $\int_{\mathbb{R}_+} e^{px}\nu(x)dx < \infty$ for p in some open interval including zero, so $\tilde{J}^\varepsilon$ has positive-only jumps. We refer to [BPS24] for details, applications and generalizations of (4).

Then setting $z = x/\varepsilon$, $dx = \varepsilon dz$, and we see that

$$\frac{1}{\varepsilon}\int_0^\infty x(N^\varepsilon(dx,dt) - V_t^\varepsilon\nu(x)dxdt) \;=\; \int_0^\infty z(N^\varepsilon(\varepsilon dz,dt) - V_t^\varepsilon\nu(\varepsilon z)\varepsilon dzdt)$$

Then (from Theorem 3 in [BPS24]), $F(\psi(u))$ in (3) will now include an additional term $V_\varepsilon(\varepsilon\psi(u))$, where $V_\varepsilon(p) := \int_0^\infty (e^{pz} - 1 - pz)\varepsilon\nu(\varepsilon z)dz$ is the log mgf of the re-scaled Lévy component, so

$$V_\varepsilon(\varepsilon p) \;=\; \int_0^\infty (e^{p\varepsilon z} - 1 - \varepsilon pz)\varepsilon\nu(\varepsilon z)dz \;=\; \int_0^\infty (e^{px} - 1 - px)\nu(x)dx \;=\; V_1(p) \tag{5}$$

and $V_1^{(n)}(0) = \int_0^\infty x^n\nu(x)dx < \infty$ for $n \geq 2$, so V_1 is still analytic at zero, so (again from Lemma 4.4 in [FGS21]), $\frac{1}{\varepsilon}\phi_\varepsilon(t)$ tends to $U_J(p)$ which is the smallest root of $F_J(u) = \frac{1}{2}(p^2 - p) + (\rho p\sigma - \lambda)u + \frac{1}{2}\sigma^2u^2 + V_1(u)$ (note V_1 and F_J are convex, see e.g. Lemma 2.2.5 in [DZ98]).

In the next subsection we construct a time-changed arithmetic Brownian motion whose marginal at time 1 has log mgf $U_J(p)$.

2.2 Characterizing the limiting law for X^ε in terms of a Lévy process

Let Z be a spectrally positive Lévy process with Lévy triple (b, σ^2, ν) with $b = \rho p\sigma - \lambda$ and log mgf $\Lambda(u) = (\rho p\sigma - \lambda)u + \frac{1}{2}\sigma^2u^2 + V_1(u)$ at $t = 1$. Then from the optional stopping theorem

$$\mathbb{E}(e^{-qH_a}) \;=\; e^{-\Phi(q)a} \;=\; e^{\Phi(q)|a|} \tag{6}$$

for $q \geq 0$ and $a < 0$ where $H_a = \inf\{t \geq 0 : Z_t < a\}$ and $\Phi(q) \leq 0$ is smallest root of $\Lambda(u) = q$ (since we have a *lower* barrier here), see e.g. Eq 2.5 in [KKR13] or Appendix C in [BF25] for details and the limiting argument using the dominated convergence theorem [2], and (6) also holds for $q < 0$ if q is greater than the minimum value attained by $\Lambda(.)$. add Sato reference etc.

If we now let $W_t^{(-\frac{1}{2})} = W_t^\perp - \frac{1}{2}t$ where $W^\perp$ is another Brownian motion independent of Z and everything else, then for p in some non-zero interval around zero, we see that

$$\mathbb{E}(e^{pW_{H_{-t}}^{(-\frac{1}{2})}}) \;=\; \mathbb{E}(e^{\frac{1}{2}(p^2-p)H_{-t}}) \;=\; e^{t\Phi_p(-\frac{1}{2}(p^2-p))} \tag{7}$$

and $\Phi_p(-\frac{1}{2}(p^2 - p))$ is $U_J(p)$ at the end of the previous subsection (note that H_{-t} also depends on p).

Then it turns out that (7) is the mgf of $\eta_{\tau_t^-}$, where $\eta_t = W_t - \frac{1}{2}t$ and $\tau_t^- = \inf\{s : \xi_s < -t\}$, where $\xi_t = -\lambda t + \sigma(\rho W_t + \bar{\rho}\tilde{W}_t) + J_t$, where W and $\tilde{W}$ are independent and J is a spectrally positive Lévy process (independent of $(W, \tilde{W})$), with $\log\mathbb{E}(e^{uJ_1}) = V_1(u)$.

To see this, we note that the joint mgf of (ξ_1, η_1) is

$$\Psi(u,p) \;=\; \log\mathbb{E}(e^{u\xi_1 + p\eta_1}) \;=\; \Lambda_p(u) + \frac{1}{2}(p^2 - p)$$

with $\Lambda_p(u) = -\lambda u + \frac{1}{2}\sigma^2u^2 + V_1(u) + \rho\sigma pu$. Now define $\kappa(p)$ by the branch through zero:

$$\Lambda_p(\kappa(p)) + \frac{1}{2}(p^2 - p) \;=\; 0$$

[2] in [KKR13] they take the largest root since they are considering an upper barrier

with $\kappa(0) = 0$. Then $\kappa(p) = \Phi_p(-\frac{1}{2}(p^2 - p))$, and

$$\mathbb{E}(e^{\kappa(p)\xi_{\tau_t^-} + p\eta_{\tau_t^-}}) \quad = \quad \mathbb{E}(e^{-\kappa(p)t + p\eta_{\tau_t^-}}) \quad = \quad 1$$

so $\mathbb{E}(e^{p\eta_{\tau_t^-}}) = e^{\kappa(p)t}$ assuming $\tau_t^- < \infty$ a.s., which is true because $\Lambda_0'(0+) = -\lambda < 0$, where $\Lambda_0(u) = -\lambda u + \frac{1}{2}\sigma^2 u^2 + V_1(u)$ and $V_1'(0) < \infty$.

Thus we have shown the following:

Proposition 2.2 *X_t^ε tends weakly to X_t as $\varepsilon \to 0$, where X_t has the marginal law of $\eta_{\tau_t^-}$.*

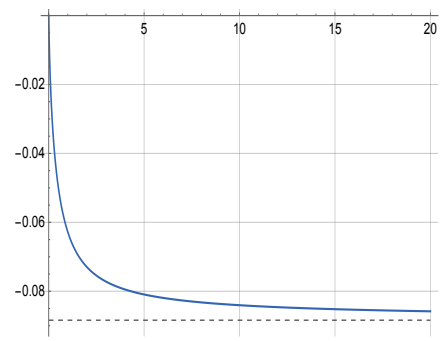

Figure 1: Here we see convergence of the VIE $\psi(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} F(\psi(s))ds$ to the asymptotic solution $U_J(p)$ for $H = .2$, $\nu = 1$; $\lambda = 1$, $\rho = -.65$, $p = \frac{1}{2}$ and $\nu(x) = \frac{e^{-Mx}}{x^{1+Y}} 1_{x>0}$ with $M = 2$ and $Y = 1.5$ using an Adams scheme with 2000 time steps.

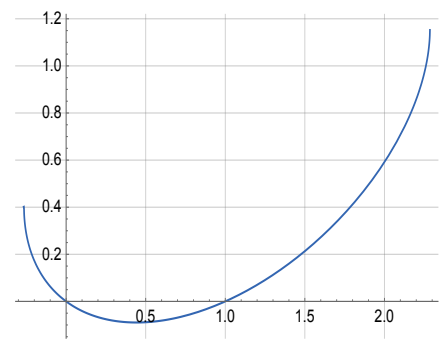

Figure 2: The log mgf $U_J(p)$ of $\eta_{\tau_t^-}$ at $t = 1$ for the same parameters in the plot above. $U_J(p) = \infty$ for p outside the domain shown in the plot.

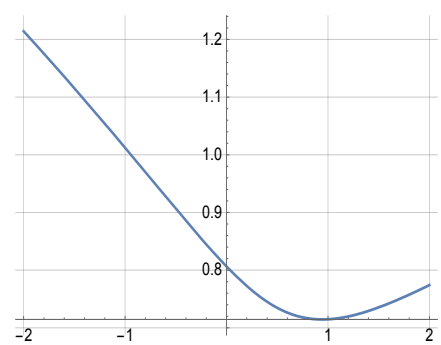

Figure 3: Here we have plotted the implied volatility smile as a function of log-moneyness $x = \log \frac{S_0}{K}$ for the asymptotic model $S_t = e^{\eta_{\tau_t^-}}$ in Proposition 2.2 for the same parameters as above but with $C_1 = .001$ and $\rho = -.2$. Note this asymptotic smile does not depend on V_0 or H.

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